

Robo Soccer Robot

Product Requirements Document

Parametric concept, interfaces, and CAD handoff specification

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Design status: concept-level, dimensioned for CAD handoff.

Fabrication status: do not manufacture until all placeholder hardware dimensions are measured.

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1 Executive summary

This document defines a compact four-wheel-drive robo-soccer platform based on a 200 mm × 200 mm square-tube chassis, 65 mm wheels, four geared motors, four BTS7960 motor drivers, an ESP32-class controller, a 12 V battery, a radio receiver, and a front ball-contact bracket. The design prioritizes a low center of gravity, serviceability, inboard wheels, and a footprint that remains below the 250 mm × 250 mm competition envelope.

The accompanying CAD sources include an editable CadQuery generator and an exported STEP assembly. The assembly shows the major parts, the mechanical stack-up, electronics envelopes, and the major service interfaces. The source intentionally distinguishes **fixed**, **derived**, and **interface-parameterized** dimensions so that a later CAD pass can replace manufacturer-specific details without silently changing the design intent.

The first engineering gate is dimensional verification: measure the actual tube, wheel, motor gearbox, battery, driver heatsink, receiver, and controller before releasing fabrication drawings.

2 Product definition

2.1 Objective

Build a robust, serviceable robot that can drive, turn, and interact with a soccer ball inside a maximum 250 mm × 250 mm footprint. The robot should keep heavy components low, protect wiring from wheel motion, and expose enough access for rapid battery and electronics service.

2.2 Intended use

- Indoor or smooth-field robo-soccer prototype.
- Four-wheel drive with independent motor control.
- Front ball-contact bracket with a nominal 70 mm opening.
- Manual radio control in the first integration phase; autonomous strategy is a future extension.

2.3 Success criteria

ID	Target	Acceptance evidence
PR-01 Footprint	$\leq 250 \times 250$ mm	Measured assembled envelope including wheels and bracket
PR-02 Chassis	200 × 200 mm	CAD inspection and tape-measure check
PR-03 Drive	4 independently driven wheels	Bench test of all four channels
PR-04 Serviceability	Battery removable without tools	Remove/install demonstration
PR-05 Stability	CG height ≤ 55 mm target	Mass-property estimate after hardware selection

PR-06 Ball interface	70 mm nominal opening	Ball-contact fit check
PR-07 Electrical safety	Upstream disconnect	Continuity and fault-isolation test

3 System architecture

3.1 Coordinate system and design intent

The CAD origin is at the chassis center on the floor-pan reference plane. +X points forward toward the ball bracket, +Y points to the left side of the robot, and +Z points upward. All dimensions in this document and in the CAD source are millimetres unless stated otherwise.

The primary layout is symmetric about both center planes. The two 200 mm longitudinal rails sit at $Y = \pm 90$ mm. Two 160 mm cross rails sit at $X = \pm 90$ mm, inside the longitudinal rails. The resulting outer structural envelope is 200 mm $\times$ 200 mm.

3.2 Functional blocks

Block	Function	Interfaces
Energy	Store and distribute 12 V power	Battery, fuse/disconnect, motor-driver power, buck converter
Drive	Convert commands into wheel torque	ESP32 PWM/DIR $\rightarrow$ BTS7960 $\rightarrow$ four motors
Control	Run manual-control firmware and future strategy	ESP32, receiver input, serial/programming header
Motion	Translate wheel torque into field motion	Four wheels, four motor mounts, floor clearance
Ball	Guide/contact the ball at the front	70 mm opening, low capture bar, replaceable liner

4 Mechanical requirements

4.1 Global parameters

Parameter	Value	Status	Engineering note
FOOTPRINT_MAX	250 $\times$ 250	Fixed	Competition envelope limit
CHASSIS_L / CHASSIS_W	200 / 200	Fixed	Square-tube outer structure
TUBE	20 $\times$ 20	Placeholder	Measure actual PVC tube; model currently assumes 20
WHEEL_D	65	Fixed	Wheel diameter used for axle height

WHEEL_W	25	Fixed by brief	Verify hub protrusion and bore before machining
EDGE_CLEARANCE	15	Fixed target	Symmetric inboard wheel placement
MOTOR_D / MOTOR_L	37 / 63	Fixed envelope	37GB-family envelope; gearbox hole pattern remains editable
MOTOR_SHAFT_D / L	6 / 15	Placeholder	D-shaft flat orientation matters
BATT_L / W / H	150 / 45 / 45	Fixed envelope	12 V, 7000 mAh envelope from brief; verify connector exit
BTS_L / W / H	50 / 50 / 40	Fixed envelope	Simplified board/heatsink solid; verify exact mounting pattern
GROUND_CLEARANCE	≈ 22	Derived	65/2 minus approximately 10 mm chassis drop

4.2 Chassis and drivetrain geometry

- Wheel centers are at $(X, Y) = (\pm 72.5, \pm 72.5)$ mm.
- Track and wheelbase are both 145 mm.
- Wheel axles are nominally collinear at $Z = 32.5$ mm.
- Wheels are inboard. Outboard wheels would push the envelope to approximately 255 mm across and violate the 250 mm limit.
- Add a 5 mm relief slot in each tube wall at the wheel well after the real wheel and tube geometry are measured.
- Heavy components sit on or near the floor pan. Target CG height is no more than 55 mm.

4.3 Motor mount requirements

Each corner uses a mirrored L-bracket made from 3 mm aluminium or printed PETG. The bracket shall provide:

1. A motor face plate with four M3 holes on the measured gearbox pattern.
2. Two vertical-leg M3 adjustment slots allowing approximately ± 2 mm axle-height correction.
3. Two M3 through-bolts into the tube using T-nuts or captive nuts.
4. A mount datum that keeps all four axles collinear after adjustment.
5. A replaceable interface between the gearbox and the bracket so motor replacement does not require frame disassembly.

5 Electronics and wiring

5.1 Placement

The battery is centered at $X = 0, Y = 0$ on the floor pan. It is held with hook-and-loop and two removable retainers bolted to the cross rails. Four BTS7960 drivers occupy open positions around the battery at nominal centers $(\pm 60, 0)$ and $(0, \pm 60)$, each on four 10 mm M3 standoffs. This leaves airflow around the heatsinks and keeps high-current motor wiring short.

The ESP32 board, buck converter, and receiver form the controller stack near the front/side service area. The CAD model shows their nominal envelopes, not their exact PCB hole patterns; verify the selected board variants before drilling.

5.2 Electrical interface contract

Interface	From	To	Requirement
BAT+ / BAT-	12 V battery	Fuse/disconnect and power bus	Use a single upstream disconnect; common ground throughout
MOTOR_PWR	12 V bus	BTS7960 × 4	Route high-current pairs away from receiver and signal wires
MOTOR_OUT × 4	BTS7960	Gear motors	One driver per motor; label left/right and front/rear
PWM/DIR × 4	ESP32	BTS7960 inputs	Use four independent channels; reserve GPIOs in firmware map
5 V LOGIC	Buck converter	ESP32 + receiver	Verify regulator current margin and connector polarity
RC SIGNAL	Receiver	ESP32	Define PWM/PPM/serial protocol before harness fabrication
DEBUG/PROG	USB/UART header	ESP32	Accessible without removing battery

5.3 Wiring and safety requirements

- Add an appropriately sized fuse or resettable protection upstream of the motor-driver bus.
- Use a keyed or polarized connector for the battery and a physically protected power switch/disconnect.
- Separate high-current motor pairs from receiver antenna and controller signal wiring.
- Provide strain relief at every connector that leaves the floor pan.
- Label all four motor channels at both ends of the harness.
- Confirm motor-driver logic voltage compatibility with the selected ESP32 board.
- Provide a single-point common ground and verify it with a continuity test before powering the controller.

6 Software and control requirements

6.1 Minimum control behavior

The first firmware milestone shall support:

- Arm/disarm state with an explicit safe default after boot.
- Independent output for four motor channels.
- Radio-loss timeout that commands all motor outputs to zero.
- Battery-voltage telemetry or at least a low-voltage warning input in the architecture.
- Direction and inversion parameters per wheel so wiring orientation is configurable.
- A test mode that spins one motor at a time at low duty cycle.

6.2 Future extension points

Reserve software and wiring capacity for wheel-speed sensing, a camera or range sensor, and autonomous ball strategy. These are intentionally not modeled as purchased hardware in the first CAD release.

7 CAD deliverable specification

7.1 Included source model

The deliverable includes `robo_soccer_bot_assembly.step`, a real multi-solid STEP assembly that re-imports successfully through a STEP-capable CAD kernel, plus `robo_soccer_cad.py`, its editable CadQuery generator. The earlier OpenSCAD source remains available as a lightweight parametric reference. The CAD deliverables include:

- Frame rails, floor pan, and service deck.
- Four wheels with hubs and four motor pods.
- Mirrored L-bracket motor mounts with fastener markers.
- Battery envelope and removable retainer rails.
- Four BTS7960 envelopes with standoff markers.
- ESP32, buck, and radio-receiver envelopes.
- Front ball bracket, opening, and low capture bar.
- EXPLODED, SHOW_WHEELS, SHOW_ELECTRONICS, and SHOW_FASTENERS toggles.

Open the STEP file in FreeCAD, Fusion 360, SolidWorks, or another STEP-capable CAD tool. The native CadQuery generator exposes editable interface parameters, including the motor gearbox hole spacing. The accompanying SVG files provide CAD-derived top and isometric views.

7.2 Assembly sequence

1. Cut and deburr the two longitudinal and two cross rails.
2. Assemble the square frame and verify 200 mm outer dimensions.
3. Install the floor pan and check that it does not interfere with wheel sweep.
4. Fit the four motor brackets loosely, set axle height, and verify collinearity.
5. Install motors and wheels; rotate by hand through a full revolution.
6. Install battery retainers and confirm tool-free battery removal.
7. Install the four drivers on standoffs with open airflow paths.
8. Install controller stack, receiver, and buck converter.
9. Route and label the harness, then perform continuity checks before energizing.
10. Fit the ball bracket and verify the final footprint remains within 250 mm × 250 mm.

8 Verification and test plan

ID	Test	Method	Pass condition
V-01	Envelope	Measure max X/Y including wheels and bracket	Both $\leq$ 250 mm
V-02	Wheel clearance	Rotate each wheel by hand with chassis mounted	No rub or hard spot

V-03	Axle alignment	Straightedge or datum measurement	All axle centers collinear
V-04	Battery service	Remove/install without tools	No frame disassembly
V-05	Driver cooling	Low-speed and sustained bench run	No unsafe heat rise; airflow clear
V-06	Electrical isolation	Continuity, polarity, and disconnect test	No short; disconnect removes drive power
V-07	Channel mapping	Run one motor at a time	Correct direction and label
V-08	Ball interface	Fit intended ball and push against bracket	Stable contact without wheel interference

9 Risks, assumptions, and open decisions

9.1 High-priority risks

- Actual battery connector and cable-exit dimensions may affect service access.
- The selected motor gearbox hole pattern or shaft geometry may differ from the 37GB-family envelope; the interface is intentionally parameterized.
- BTS7960 mounting details may differ from the simplified 50 × 50 × 40 mm solid.
- The front ball bracket may change the footprint if its liner or capture geometry projects beyond the 250 mm limit.
- PVC tube wall thickness and fastener access may require local reliefs not represented in this concept model.

9.2 Open decisions before release to manufacture

1. Confirm the exact motor, wheel, battery, driver, ESP32, buck, and receiver part numbers.
2. Confirm manufacturer-specific interfaces and update the CAD parameters where required.
3. Choose the final tube material and joining method: printed corner blocks, aluminium angle, or direct fasteners.
4. Select fuse rating, wire gauges, connectors, and an emergency disconnect.
5. Confirm field rules for ball size, robot envelope, and allowed materials.
6. Decide whether a protective top cover is required after the first drive test.

10 Release checklist

- ☐ Manufacturer-specific interfaces verified and editable parameters updated.
- ☐ CAD assembly opens and exports without geometry collisions.
- ☐ Footprint and wheel clearance verified on the assembled prototype.
- ☐ Electrical harness labeled, fused, and polarity-checked.
- ☐ Safe boot, radio-loss timeout, and one-motor-at-a-time test completed.
- ☐ Ball bracket and service access approved by the build owner.

11 Appendix: parameter handoff

Coordinate system: X forward, Y left/right, Z up

Fixed: CHASSIS_L=200, CHASSIS_W=200, FOOTPRINT_MAX=250, WHEEL_D=65

Derived: WHEEL_X=72.5, WHEEL_Y=72.5, AXLE_Z=32.5

Interfaces to verify: TUBE, wheel hub/bore, gearbox hole pattern, shaft D/L, battery connector exit, BTS mounting pattern, ESP32 variant, buck, receiver

Interfaces: 12V power bus; 4x motor outputs; 4x PWM/DIR;

5V logic; receiver signal; USB/UART debug; upstream disconnect

End of document — concept design release pending hardware measurement